

List of Repair Groups

General	Technical Data	Page 0.1
Repair Groups		Group
	Maintenance, Self-diagnosis	03
Engine	Engine, Crankcase	10
	Engine, Crankshaft, Pistons	13
	Engine, Cylinder Head and Valve Drive	15
	Engine, Lubrication	17
	Engine, Cooling	19
	Fuel Supply	20
	Air Flow Controlled Fuel Injection	24
	Exhaust System/Emission Controls	26
	Starter, Power Supply, Cruise Control	27
	Ignition System	28
Transmission	Clutch, Controls	30
	Torque Converter	32
	Manual Transmission, Controls, Case	34
	Manual Transmission, Gears, Shafts	35
	Automatic Transmission, Controls, Case	37
	Automatic Transmission, Gears, Valve Body	38
	Differential, Transaxle System	39
Chassis	Front Wheel Suspension	40
	Rear Wheel Suspension, Axle Shaft	42
	Wheels, Tires, Alignment	44
	Antiblock System	45
	Brakes, Mechanical	46
	Brakes, Hydraulics	47
	Steering	48
Body	Body-Front Section	50
	Body-Center Section	51
	Body-Rear Section	53
	Lids	55
	Doors	57
	Hardtop	61
	Bumpers	63
	Glasses, Window Control	64
	Exterior Equipment	66
	Interior Equipment	68
	Seats	72
	Seat Covers	74
Heating,	Heater	80
Ventilation,	Ventilation	85
Air Condition	Air Conditioner	87
Electrics	Instruments, Fuel Gauge, Alarm System	90
	Radio	91
	Windshield Wipers and Washer	92
	Exterior Lights, Lamps, Switches	94
	Interior Lights	96
	Wiring	97

	Page
General	
International Unit System	.0.01
Lifting Car	.0.1
Technical Data, General	.0.3
Technical data - Type 928 GT - 89 model	.0.35
Technical Data - Type 928 GTS - 1992 Model Year	.0.36
Maintenance, Self-diagnosis	
LH/EZK control unit error diagnosis	.03-1
Connection in the 928 S 4	.03-3
Starting error diagnosis	.03-5
Functional check of actuator and input signals	.03-7
Test code list	.03-13
System adaption	.03-15
Knock detection	.03-17
Trouble-shooting	.03-19
Error code list for the LH control unit	.03-27
Trouble-shooting, EZK control unit	.03-29
Error code list for the EZK control unit	.03-37
Resetting the error memory	.03-39
Operating instructions for system tester 9288	.03-41
Engine / Crankcase	
Tolerances and wear limits	.10-01
Technical data	.10-02a
Engine torque specifications	.10-03
Removing and installing engine	.10-1
Engine, removing and installing	.10-11
Engine, Crankshaft, Pistons	
Tools	.13-1
Engine support (special tool 9188)	.13-2
Crankcase tools	.13-3
Disassembling and assembling crankcase and crankshaft	.13-4
Sealing upper and lower crankcase sections	.13-7
Checking crankshaft bearing clearance	.13-8
Installing crankshaft oil seals	.13-9
Installing upper and lower crankcase sections	.13-10
Machining flywheel	.13-10b
Machining flywheel - 1984 models	.13-10b

Page

Removing and installing flywheel	13-11
Removing and installing pilot bearing	13-11
Disassembling and assembling pistons and connecting rods	13-13
Installing piston and connecting rod	13-15
Tolerance group of pistons and cylinders	13-16
Checking connecting rod bearing clearance	13-16
Replacing belts for air pump, alternator and power steering pump	13-17
Checking and adjusting belt tightness	13-18
Checking and adjusting tension of alternator poly-rib-belt	13-18b
Replacing drive belts for alternator, power pump air, pump and A/C compressor	13-18c
Crankshaft tools	13-19
Installing flywheel stop	13-20
Crankshaft journal sizes	13-21
Checking piston and cylinder bore	13-23
Piston arrangement	13-24a
Piston sizes, 84/85/86 models onward	13-24b
Notes for installing pistons (32 valve engines)	13-24c
Pistons, 87 models onward	13-24c
Checking pistons and cylinder bores (32-valve engines)	13-24d
Checking pistons and cylinder bores - Type 928 GTS (5,4 l)	13 - 24e
Remachining cylinder bores	13-25

Engine, cylinder head, valves

Valve tools	15-1
Disassembling and assembling valve drive	15-3
Checking end play of camshaft	15-6f
Camshaft survey	15-6g
Disassembling and assembling valve train	15-7
Disassembling and assembling valve drive	15-8
Machining valve seats	15-8a
Checking valve guides	15-8a
Removing and installing valve springs	15-8b
Checking and installing length of valve springs	15-8b
Removing and installing camshaft drive belt and water pump	15-9
Installing camshaft drive belt and adjusting timing	15-14
Drive belts with HTD teeth (beginning with 1983 models)	15-16
Checking and adjusting drive belt	15-17
Machining cylinder head	15-19
Disassembling and assembling tensioning roller housing	15-20
Bleeding tensioning roller housing (from 1983 models)	15-23
Disassembling and assembling tensioning roller housing	15-24

	Page
Cylinder head tools	15-27
Replacing valve guides	15-28
Checking TDC marks on camshaft sprockets	15-31
Tools (32 valve engine)	15-101
Checking and adjusting toothed belt	15-102
Replacing toothed belt	15-105
Tools - cylinder head (32 valve engines)	15-109
Disassembling and assembling cylinder head	15-110
Tools - cylinder head (32-valve engines)	15-112
Tools - cylinder head, removing and installing valve springs	15-112a
Removing and installing valve springs	15-112b
Removing and installing valve stem seal	15-112b
Installing valve stem seal, '87 models onward	15-112c
Checking valve guides	15-112d
Cylinder head - Tools replacing valve guides	15-112e
Machining cylinder-head mating face	15-112j
Checking valve seat wear limit	15-113
Checking and machining valve seats	15-114
Valve dimensions 928 S '85/'86 models onward	15-115
Checking and adjusting installation length of valve springs (32-valve engines)	15-116
Tools-Installing cylinder head	15-116d
Modifications to the cylinder head (32 valve)	15-116e
Cylinder head, installing	15-117
Tools - Disassembling and assembling valve drive	15-119
Disassembling and assembling valve drive	15-120
Camshaft installation overview	15-124
Camshaft installation overview Type 928 S4, 928 GTS (5,4 l)	15-124a
Removing and installing camshafts	15-125
Installing camshaft seal	15-128a
Removing and installing chain tensioner for camshafts	15-129
Camshaft setting, checking and adjusting	15-131
Cylinder head cover, dismantling and reassembling	15-137
928 S4 Club Sport-Version	15-139
Camshaft installation	15-140
Checking and/or adjusting camshaft settings	15-141
Checking and/or adjusting camshaft settings, Type 928 GTS (5,4 l)	15-142

Page

Engine, Lubrication

Removing and installing lubricating system parts	17-1
Oil pump, removing and installing	17-6
Replacing shaft seal for oil pump	17-6a
Lubrication system components, removing and installing	17-6b
Changing engine oil and engine oil filter	17-7
Checking oil pressure	17-8
Checking oil pressure	17-9
Cleaning oil check valve for hydraulic valve tappets	17-10
Cleaning of engine-oil system	17-11

Engine, Cooling

Changing coolant and bleeding cooling system	19-1
Changing coolant and bleeding cooling system	19-2
Checking coolant system cap	19-2a
Checking coolant thermostat	19-2b
Disassembling and assembling visco fan	19-3
Disassembling and assembling thermostat housing for cooling system (ATF)	19-6
Disassembling and assembling thermostat housing for cooling system	19-8
Disassembling and assembling thermostat housing for cooling system - Notes	19-10
Cooling-air flap control, '87 models onward	19-11
Tightening torques, overview	19-11
Function charts	19-15
Removing and installing cooling-air flap motor	19-19
Removing and installing fans	19-20
Removing and installing cooling-air flaps	19-21
Troubleshooting	19-22
Function-testing senders	19-23
Function-testing flap positioning motor	19-24
Function-testing the output stage	19-25

	Page
Fuel supply	
Replacing fuel filter - up to 1980 models	20-1
Removing and installing fuel pump	20-2
Checking delivery rate of fuel pump - CIS	20-3
Checking fuel pump delivery - AFC	20-4
Checking fuel pump delivery - LH-Jetronic, beginning with 1984 models	20-5
Hose diagram for CIS	20-6
Arrangement of check valves	20-8
Replacing check valve	20-9
Disassembling and assembling accelerator pedal	20-10
Removing and installing fuel tank	20-12
Removing and installing fuel intank (Japan)	20-14
Fuel level sensor, removing and installing	20-15
Activated charcoal cannister for removal of tonix substances	20-17
form fuel tank breather (diagram)	
Calibrating the tank sensor system for model 89 onwards	20-19
Electronic fuel injection	
AFC equipment table	24-01
Testing and adjusting specifications	24-1
Adjusting idle - USA and Japan	24-3
Adjusting idle speed and CO - USA and Japan, 1983 models	24-6a
Checking fuel pressure	24-27
Removing and installing pressure regulator	24-9
Removing and installing intake air sensor	24-9
Removing and installing cold start valve	24-10
Removing and installing auxiliary air regulator	24-10
Removing and installing temperature sensor II	24-11
Removing and installing pressure damper	24-11
Removing and installing fuel injectors	24-12
Disassembling / assembling fuel injector	24-13
Removing and installing fuel injectors	24-13
Removing and installing throttle housing	24-14
Checking and adjusting microswitches on throttle housing	24-15
Checking temperature sensor II	24-17
Checking temperature time switch (for cold start valve)	24-17
Checking auxiliary air regulator	24-18
Checking cold start valve	24-18
Checking vacuum control	24-19
Checking pressure regulator	24-19
Checking air flow sensor	24-20
Checking pressure regulator	24-20

Page

Troubleshooting AFC	24-21
Hose Diagram for AFC	24-22
Correcting CO level on air flow sensor	24-24
LH-Jetronic equipment table	24-101
LH-Jetronic air flow	24-102
LH-Jetronic	24-104
Troubleshooting LH-Jetronic	24-108
Testing oxygen sensor operation	24-120a
Testing and adjusting specifications	24-121
Adjusting idle and CO, 1984 models	24-122
Removing and installing heated wire air flow sensor	24-125
Removing and installing fuel cooler	24-126
Specifications for testing and adjusting	24-201
Equipment chart - LH-Jetronic	24-202
Adjusting idle	24-203
Correcting CO adjustment on air volume sensor	24-207
Fuel path - LH-Jetronic	24-208
Injection valves, removing and installing	24-209
Testing and adjustment values for Type 928 S4 - 928 GT - 928 GTS (5,4 l)	24-211
Equipment chart - LH-Jetronic	24-212
Checking idle, USA cars and cars with catalytic converters	24-213
Adjusting idle without catalytic converter	24-214
Fuel path, LH-Jetronic	24-216
Modified breather for crankcase and cylinder heads	24-218
Testing oxygen sensor operation	24-219
Testing oxygen sensor operatio	24-221
Coding for LH control unit	24-222
Removing and installing rotary idle actuator	24-223

Fuel mixture

Tightening torque for fuel system	25-01
Replacing air cleaner element	25-1
Adjusting idle and CO	25-2
Testing and adjusting specifications	25-5
Pressure tests	25-7
Connectig and bleeding pressure gauge	25-8
Checking system pressure	25-10
Checking control pressure "warm"	25-10
Checking entire fuel system for leaks	25-11
Checking control pressure "cold"	25-12

	Page
Checking function of hot start valve (CIS)	25-12b
Checking hot start valve for leaks (CIS)	25-12b
Pressure tests with VW 1318	25-12c
Connecting and bleeding pressure tester VW 1318	25-12d
Connecting and bleeding pressure tester VW 1318	25-12e
Checking system pressure	25-12f
Checking control pressure "warm"	25-12f
Checking and adjusting rest position of air flow sensor plate	25-13
Replacing pressure relief and residual pressure valves	25-14
Removing and installing mixture control unit	25-15
Disassembling and assembling mixture control unit	25-17
Disassembling and assembling mixture control unit	25-21
Removing and installing fuel injectors	25-23
Removing and installing oxygen sensor	25-25
Resetting elapsed mileage switch	25-26
Removing and installing control unit for AFC fuel injection and oxygen sensors	25-26

Exhaust system/ Emission Controls

Replacing filter element of air pump	26-1
Checking tightness of exhaust flanges	26-1
Hose Layout	26-3
Intermediate and/or final mufflers - single pipe system	26-6
Replacing intermediate and/or final mufflers	26-7
Intermediate and/or final mufflers - twin pipe system	26-8
Replacing intermediate muffler and final muffler	26-9
Replacing final muffler	26-9

Starter, Power Supply, Cruise Control

Checking battery	27-1
Automatic cruise control (tempostat)	27-3
Removing and installing tempostat control	27-6
Removing and installing tempostat switch (steering column switch)	27-6
Removing and installing clutch pedal switch	27-7
Removing and installing vacuum servo	27-8
Removing and installing cable on vacuum servo	27-9

	Page
Troubleshooting automatic cruise control (tempostat)	27-10
Checking multiple pin plug of control with voltmeter or test lamp	27-12
Checking multiple pin plug of control with ohmmeter and ignition turned off	27-13
Removing and installing starter	27-15
Removing and installing tempostat control unit	27-17
Removing and installing steering column switch, actuator	27-18
Removing and installing bowden cable	27-19
Removing and installing clutch pedal switch	27-20
Troubleshooting, tempostat	27-21
Tempostat control unit pin assignment	27-21
Testing multi-pin plug of tempostat control unit	27-22
Alternator, removing and installing	27-25

Ignition system

Ignition components	28-1
Cautions when repairing electronic ignition systems	28-3
Adjusting ignition timing	28-4
Distributor specifications	28-7
Removing and installing distributor	28-8
Replacing spark plugs	28-9
Removing and installing vacuum unit	28-10
Troubleshooting electronic ignition system	28-11
Ignition components	28-21
Ignition adjusting and testing values	28-22
Adjusting ignition timing	28-23
Checking ignition timing	28-24
Checking distributor	28-25
Equipment table	28-31
Checking distributor	28-34
Equipment table	28-35
cautions when repairing electronic igniton systems	28-39
Electronic ignition system	28-40
Important car information	28-44
Removing and installing electronic ignition control unit	28-45
Removing and installing transistorized ignition control unit	28-45
Removing and installing transistorized igniton control unit	28-46

	Page
Ignition System	
Removing and installing transistorized igniton control unit	28-46
Removing and installing speed/reference mark sensor	28-46
Removing and installing distributor	28-47
Removing and installing TDC sensor	28-48
Removing and installing TDC sender	48-48c
Troubleshooting electronic ignition system	28-49
Equipment survey	28-65
Replacing spark plugs	28-67
EIK control-unit coding	28-69
Checking the transmission protection switch (automatic transmission)	28-71
Equipment table	28-73
Equipment table Type 928 GT - 928 GTS (5,4 l)	28-74
Ignition circuit temperature sensor, removing and installing	28-75
Checking the ignition circuit monitoring relay	28-77
DME Diagnosing / Troubleshooting	D24/28-1
LH-Diagnosis / Troubleshooting	D24-11
EZK-Diagnosis / Troubleshooting	D28-1

	Page
Clutch, Controls	
Technical data	30-01
Torque specifications for manual transmission, gear shift, central tube	30-03
Technical data	30-05
Technical data (transmission type A 28.01)	30-08
Torque specifications	30-09
Technical data	30-013
Technical data (transmission type A 28.01)	30-014
Checking clutch play	30-1
Clutch operation	30-1
Adjusting clutch spring	30-1
Checking wear of clutch discs	30-2
Bleeding clutch	30-2a
Removing and installing clutch pedal	20-2b
Removing and installing clutch spring	30-2g
Removing and installing clutch	30-3
Removing and installing clutch, single-disk clutch, '87 models onward	30-6b
Disassembling and assembling clutch (diameter-centered)	30-7
Disassembling and assembling clutch (dowel pin-centered)	30-10
Disassembling and assembling clutch (diameter and dowel pin centered)	30-17
Clutch linkage versions	30-20
Ball stud versions	30-21
Replacing ball stud	30-21
Removing and installing clutch housing	30-22
Checking clutch discs	30-23
Checking clutch pressure plate	30-24
Checking clutch intermediate plate	30-24
Overhauling clutch slave cylinder	30-25
Checking clutch disc for wear	30-27
Torque converter	
Removing and installing torque converter	32-1
Manual transmission / controls, case	
Changing transmission oil	34-1
Removing and installing transmission	34-3
Adjusting transmission suspension	34-8a
Disassembling and assembling shift linkage	34-8c
Disassembling and assembling transmission	34-9
Disassembling and assembling upper case cover with reverse gear lock	34-19

Page

Manual transmission / Gears and shafts

Disassembling and assembling input shaft	35-1
Disassembling and assembling pinion shaft	35-5
Determining shim thickness "X"	35-11
Determining shim thickness "Y"	35-13
Measuring Distance "Z" between input shaft and pinion shaft	35-15
Disassembling and assembling synchronizers	35-17
Disassembling and assembling countershaft Hub/Gears	35-21

Automatic transmission / Controls, Case

Removing and installing transmission	37-1
Removing and installing selector cable	37-9
Removing and installing selector lever cover frame	37-10a
Adjusting selector cable	37-11
Cable and backup light switch operation	37-12
Removing and installing accelerator cable	37-13
Adjusting throttle pressure cable	37-14
Neutral/park starting switch	37-15
Disassembling and assembling front converter housing	37-17
Removing and installing final drive housing	37-21
Removing and installing rear transmission case and rear converter housing	37-26a
Disassembling and assembling rear converter housing	37-27
Disassembling and assembling final drive housing	37-31
Disassembling and assembling transmission	37-35
Disassembling and assembling transmission case	37-45
Installing automatic transmission section	37-53

Automatic Transmission / Gears, Valve Body

Checking transmission operation	38-1
Troubleshooting	38-11
Removing and installing valve body	38-21
Replacing ATF and filter	38-25
Flushing ATF cooler and lines	38-28
Disassembling and assembling centrifugal governor	38-29
Disassembling and assembling gear assembly	38-33
Disassembling and assembling clutch K 1	38-41
Disassembling and assembling clutch K 2	38-47

Page

Differential / Transaxle System

Removing and installing differential	39-1
Disassembling and assembling differential	39-5
Adjusting pinion and ring gear	39-11
Removing and installing final drive for automatic transmission	39-21
Disassembling and assembling bearing assembly for automatic transmission	39-25
Determining thickness of shims for bearing assembly	39-29
Adjusting pinion and ring gear (automatic transmission)	39-31
Removing and installing central tube (automatic transmission)	39-41
Checking drive shaft (automatic transmission)	39-49
Disassembling and assembling limited slip differential	39-55
Removing and installing central tube (manual transmission)	39-61
Disassembling and assembling central tube (manual transmission)	39-63
Checking drive shaft (manual transmission)	39-65

General

5 speed manual transmission type G 28	30-0201
Technical data	30-0203
Torque specifications	30-0205

Manual Transmission / Controls, Case

Disassembling and assembling transmission	34-201
Disassembling and assembling manual transmission ('87 models onward)	34-214a
Adjusting shift forks	34-215
Adjusting shifts	34-217
Adjusting gearshift ('86 models onward)	34-218a
Adjusting gearshift ('87 models onward)	34-218c
Disassembling and assembling upper transmission cover with reverse gear arrest	34-219
Disassembling and assembling upper transmission cover ('87 models onward)	34-220
Dismantling and assembling housing cover with oil pump	34-221
Replacing O-ring for spray tube (type G28/57)	34-223

Manual Transmission / Gears and Shafts

Disassembling and assembling drive shaft	35-201
Disassembling and assembling drive pinion	35-205
Disassembling and assembling countershaft	35-215
Disassembling and assembling layshaft ('87 models onward)	35-219

	Page
Differential / Transaxle System	
Removing and installing differential	39-201
Measuring the plate wear on the proportional-slip differential	39-206a
Removing and installing the proportional-slip differential	39-206c
Bleeding hydraulics of Porsche controlled slip differential (PSD)	39-206i
Disassembling and assembling differential	39-207
Limited-slip differential, disassembling and assembling (40% lock-up)	39-212a
Dismantling and assembling the proportional-slip differential	39-212i
Dismantling and assembling lim. slip. differential with Valeo friction discs	39-212s
Adjusting pinion and ring gear	39-213
Determining thickness of shim S5	39-215
Drive shaft and ring gear, adjusting	39-217
Adjusting and drive pinion and ring gear	39-219
Adjusting drive pinion and ring gear	39-221
Adjusting drive pinion and ring gear	39-225

PSD diagnosis / Troubleshooting: beginning with supplement 35, pages see volume III.

	Page
General Data	
Technical data	30-0101
Tightening torques	30-0105
Torque Converter	
Removing and installing torque converter	32-101
Removing and installing selector lever cover frame	37-101
Removing and installing selector lever cable	37-103
Adjusting selector lever cable	37-105
Removing and installing starter locking and backup light switch	37-107
Adjusting starter locking and backup light switch	37-109
Checking adjustment of selector lever cable and starter locking / backup light switch	37-111
Bowden cable for control pressure, removing and installing (service solution)	37-113
Bowden cable for control pressure, adjusting	37-115
Removing and installing transmission	37-117
Adjusting transmission suspension	37-123
Disassembling and assembling front converter housing	37-125
Disassembling and assembling front converter housing ('87 models onward)	37-130
Removing and installing rear transmission case	37-131
Removing and installing rear transmission	37-135
Disassembling and assembling transmission	37-143
Transmission / Gears, Valve Body	
Checking transmission operation	38-101
Checking operation	38-110
Checking operation (87 models onward)	38-110a
Troubleshooting	38-111
Removing and installing shift valve casing	38-123
Disassembling and assembling shift valve housing	38-125
Disassembling and assembling shift valve casing ('87 models onward)	38-130a
Removing and installing lower cover	38-131
Disassembling and assembling lower cover	38-133
Replacing ATF and filter	38-137
Flushing ATF cooler and lines	38-141
Removing and installing governor	38-143
Checking governor	38-145
Removing and installing vacuum modulator	38-147
Removing and installing primary pump	38-149

	Page
Disassembling and assembling secondary pump	38-155
Disassembling and assembling gear assembly	38-157
Disassembling and assembling one-way clutch	38-163
Disassembling and assembling clutch K 1	38-169
Disassembling and assembling clutch K 2	38-177

Differential / Transaxle system

Removing and installing final drive	39-101
Disassembling and assembling final drive	39-107
Disassembling and assembling differential	39-113
Disassembling and assembling bearing assembly	39-119
Disassembling and assembling bearing assembly / Determining thickness of shims for bearing assembly	39-123
Adjusting drive pinion and ring gear	39-125
Adjusting drive pinion and ring gear	39-127
Adjusting drive pinion and ring gear	39-128
Adjusting drive pinion and ring gear	39-130
Adjusting drive pinion and ring gear	39-133
Adjusting drive pinion and ring gear	39-135
Disassembling and assembling central tube	39-137
Checking central tube	39-139
PSD-Diagnosis / Troubleshooting	D39-201

	Page
Front wheel suspension	
Technical data	40-01
Tightening torques	40-02
Notes on front axle, 86 models onward	40-04
Disassembling and assembling wheel suspension	40-1
Disassembling and assembling wheel suspension, until end of model year 85	40-2
Disassembling and assembling wheel suspension, 86 models onward	40-7
Disassembling and assembling wheel suspension (before and after model year 86)	40-13
Wheel suspension components - installation overview	40-16
Modifications to mountings	40-19
Closing slots in ball joints	40-23
Inspecting protective caps of ball joints on control arms and track rods	40-24
Modified steering knuckle, cars with floating brake calipers	40-25
Disassembling and assembling wheel bearing	40-27
Removing and installing spring strut	40-35
Disassembling and assembling spring strut, tools	40-35
Disassembling and assembling non-adjustable spring strut	40-37
Disassembling and assembling adjustable spring strut	40-39
Disassembling and assembling adjustable and non-adjustable spring struts	40-42
Coil springs for adjustable spring struts	40-46
Coil springs for non-adjustable spring struts	40-47
Coil springs for non-adjustable struts/effect on car height	40-49
Dampers - installation overview	40-51

Rear Wheel Suspension, Axle Shaft

Technical data	42-01
Tightening torques	42-02
Disassembling and assembling rear suspension	42-1
Disassembling and assembling Hub assembly	42-7
Replacing upper control arm mount	42-13
Removing and installing spring strut	42-14
Disassembling and assembling spring strut	42-15
Damper - installation overview	42-18d
Removing and installing axle shaft	42-19
Disassembling and assembling axle shaft	42-21
Removing and installing rear axle assembly	42-29

Wheels, Tires, Alignment

Wheels and tires, pressure	44-01
17-inch wheels	44-03
Alignment settings	44-05

	Page
Sample measuring card	44-06
Alignment	44-1
Checking wheel rims	44-6
General assembly information, tires	44-7
Wheel balancing	44-10
Installing and removing wheels on car	44-19
Tire pressure warning system - General	44-21
Tire inflation pressure for vehicles with tire pressure warning system	44-26
Tire pressure warning system short test	44-28
Removing and installing pressure-operated switches (tire pressure warning system)	44-29
Removing and installing high-frequency sensor (tire pressure warning system)	44-32
Removing and installing tire pressure warning system control unit	44-33
Connector assignment of the control unit connector	44-34
Notes on trouble-shooting - Tire pressure warning system	44-35
Tire pressure warning system (RDK) diagnosis	44-301

Antiblock System

General	45-01
Design - ABS	45-03
Components - ABS	45-04
Important notes for troubleshooting and ABS test program	45-06
Important information for working on cars with ABS	45-07
Removing and installing hydraulic unit	45-1
Removing and installing electronic control unit	45-5
Removing and installing relais	45-6
Removing and installing sensors	45-8
Removing and installing speed sensor, 86 models onward	45-11

Brakes / Mechanical

General	46-01
Technical data	46-02
Technical data, 86 models onward	46-05
Torque specifications	46-08

	Page
Technical data as of MY '92	46-011
Checking thicknesss of brake pads	46-1
Removing and installing brake pads (Floating frame caliper disc brakes)	46-2
Removing and installing brake pads (floating caliper disc brakes)	46-5
Removing and installing brake pads (four-piston fixed caliper)	46-6b
Removing and installing brake pedal	46-8
Adjusting brake pressure rod (until end of model year 85)	46-9
Adjusting brake pressure rod (86 models onward)	46-10
Checking stoplamp switch setting	46-10a
Disassembling and assembling front wheel brake	46-11
Disassembling and assembling rear wheel brake	46-13
Checking brake disc lateral runout	46-16
Checking brake disc thickness	46-18
Disassembling and assembling parking brake lever	46-19
Checking and adjusting parking brake	46-21

Brakes / Hydraulics

Tightening torques	47-01
Notes on brakes, 86 models onward	47-03
Disassembling and assembling floating frame caliper	47-1
Disassembling and assembling fist caliper	47-7
Replacing dampers and slides	47-12
Removing and installing brake master cylinder	47-13
Removing and installing brake booster	47-15
Removing and installing brake pressure regulator	47-17
Braking-force regulators - installation overview (until end of model year 83)	47-18a
Braking-force regulator - installation overview (84 models onward)	47-18b
Testing braking-force regulator	47-19
Notes on bleeding brakes and changing brake fluid	47-22

Steering

Technical data	48-01
Torque specifications	48-02
Removing and installing steering gear	48-1
Steering stop versions	48-6a
Removing and installing power steering pump	48-7
Replacing seals for aluminium power steering pump	48-8a
Notes on power steering:	48-8i
Checking drive belt tension	48-8i
Checking steering system for leaks	48-8j

	Page
Bleeding steering system	48-8k
Checking hydraulic function of steering (pressure test)	48-9
Removing and installing steering wheel	48-14
Repairing steering and intermediate shafts with steering tube	48-15
Replacing upper steering shaft bearing	48-30
Dismantling and assembling steering wheel height adjuster of airbag vehicles	48-33

Heater

Heater, blower, air distribution housing	80-1
--	------

Air Conditioner

Specifications	87-01
Tightening torques	87-02
Refrigerant system layout	87-1
Vacuum system layout	87-2
Safety regulations for handling refrigerant R12	87-3
Installation work on the air-conditioning system (Service unit SECU)	87-4b
Removing and installing heater / evaporator	87-5
Removing and installing blower	87-16
Removing and installing air distribution housing	87-17
Disassembling and assembling heater / evaporator	87-19
Removing and installing condenser fan	87-24
Removing and installing receiver-drier	87-25
Removing and installing compressor	87-28
Disassembling and assembling compressor	87-31
Regulators and control unit of automatic air conditioner	87-43
Disassembling and assembling center vent and switch	87-47
Automatic air conditioner 84 models onward	87-89
Technical data	87-90
Tightening torques	87-91
Disassembling and assembling heater / air conditioner	87-92
Removing and installing heater / air conditioner	87-97
Removing and installing compressor	87-99
Removing and installing condenser	87-100
Removing and installing fluid tank	87-101
Removing and installing antifreeze unit	87-102
Removing and installing expansion valve / Distribution of oil in refrigeration circuit	87-103
Vacuum system - schema	87-104
Checking vacuum system	87-105
Flap position program	87-106
Checking flap position program	87-108
Checking operating switch	87-110
Testing setting motor	87-113

	Page
Adjusting temperature mixture flaps	87-115
Pressure and temperature specifications	87-116
Troubleshooting	87-118
Auxiliary air conditioner	87-120
Removing and installing magnetic clutch	87-126
Technical Data of air conditioning system as of MY '93	87-129

	Page
Body - General	
Special tools and shop materials for body repairs	50-01
Repair dimensions	50-03
Safety notes	50-05
Body-Front Section	
Replacing part of front wheelhouse	50-1
Replacing front end partially	50-5
Modifications on body front end	50-9
Body-Center Section	
Replacing rocker panel	51-1
Body-Rear Section	
Replacing rear apron and quarter panel	53-1
Replacing quarter panel end section	53-5
Body repairs to foam-filled cavities	53-7
Lids	
Removing and installing engine hood cable	55-1
Replacing latch on rear lid lock (upper section)	55-3
Removing and installing electric rear lid unlocking device	55-5
Doors	
Removing and installing door	57-1
Removing and installing door window and regulator	57-3
Removing and installing door lock	57-6
Checking door central locking system	57-7
Replacing door lock cylinder	57-15
Modification to rotary knob lock	57-17
Modification on unfinished shell	57-19
Troubleshooting central locking system - from 1980 models	57-21
Changes on doors and door lock cylinders	57-25
Sun roof	
Removing and installing power sun roof	60-1
Removing and installing drive motor and gearbox for sun roof	60-4
Removing and installing wind deflector for sun roof	60-5
Removing and installing gates, guide, cables and guide rails	60-6
Modifications on sun roof	60-8

Page

Bumpers

Removing and installing bumper	63-2
Removing and installing nose panel, '87 models onward	63-4a
Disassembling and assembling bumpers	63-5
Removing and installing bumper	63-6

Glases, Window Control

Removing and installing power window operating switch	64-1
Installing adhesive plate for inside rear view mirror	64-2
Bonding the interior rearview mirror in place	64-2a
Removing and installing windshield	64-3
Removing and installing the bonded windshield	64-7
Removing and installing the bonded rear side window	64-15
Removing and installing windshield - 2-pack adhesive	64-21
Removing and installing bonded rear side window - 2-pack adhesive	64-31

Exterior Equipment

Paint colors from 1978 models	66-01
Paint colors from 1980 models	66-02
Body paint colors from 1982 models	66-03
Body paint colors from 1984 models	66-04
Body paint colors from 1985 models	66-05
Body paint colors, '86 models onward	66-06
Body paint colors '87 models onward	66-07
Body paint colors - 1989 models onward	66-08
Body paint colors - 1990 models onward	66-09
Body paint colors - 1991 models onward	66-010
Body paint colors - 1992 models onward	66-011
Removing and installing side guard strips	66-1
Service installing side guard strips	66-3
Removing and installing front spoiler	66-5
Removing and installing rear spoiler	66-9
Removing and installing outside mirror	66-13
Service installing outside mirror on passenger's door	66-15
Removing and installing sill covers	66-20
Removing and installing rear spoiler	66-25
Removing and installing front air dam	66-27
Fitting the CLUB SPORT emblem	66-29
Retrofitting new mirror generation for vehicles as of MY '87	66-31

	Page
Interior Equipment	
Removing and installing rear side trim panel	68-1
Safety regulations for handling airbag vehicles	68-3
Removing and installing the airbag steering wheel	68-5
Removing and installing airbag components	68-6
Checking the airbag system to ensure that long-term functioning is guaranteed	68-11
Checking the safety belts	68-13
Correct disposal of airbag units	68-15
Diagnosis/Troubleshooting Airbag	D68 - 1
Seats	
Removing and installing power seat	72-1
Removing seat backrest motor	72-3
Disassembling seat controls	72-4
Disassembling left seat track and removing fore / aft seat control motor	72-5
Disassembling right seat track and removing seat height control motor	72-6
Seat and mirror position control system	72-7
Seat and mirror position control system, method of operation	72-9
Seat control unit, plug assignment	72-11
Checking seat and mirror position control system	72-13
Plug assignment, mirror control unit	72-17
Checking mirror control unit	72-19
Checking control panel	72-22
Electrically adjustable seat	72-23
Calibrating controllable seat heating	72-31
Checking seat heating	72-33
Instruments, Fuel Gauge, Alarm System	
Precautions when working on cars with electronic control units	90-01
Location of relays and electronic controls in car	90-1
Removing and installing instrument cluster	90-7
Disassembling and assembling instrument cluster	90-8
Removing and installing clock	90-10
Troubleshooting car radio	90-11
Measuring antenna resistance	90-17
Removing trim plate for radio opening	90-18
Instrument cluster as from 1979 models	90-19
Location of fuses and relays in car - 1980 model	90-20
Removing and installing radio (digital radio Bamberg QTS)	90-25
Checking fuel consumption indicator	90-27
Troubleshooting central warning system	90-29
Troubleshooting central warning system / Function test on central warning unit	90-31

	Page
Central warning unit connection plan black plug	90-32
Central warning unit connection plan yellow plug	90-33
Single component function test	90-34
Alarm system 928	90-41
Service installing Blaupunkt SQR 22 radio	90-49
Upshift indicator	90-53
Checking burglar alarm	90-55
Check auxiliary control unit - burglar alarm	90-56
Backlit instrument cluster, removing and installing	90-57
Advancing the odometer reading of the backlight instrument cluster	90-59

Radio, Telephone, Bordcomputer

Facilities for subsequent telephone installation	91-1
Retrofitting CD player from model 89 onward	91-11

Windshield Wipers and Washer

Adjusting headlight cleaner jets	92-1
Service installing rear window wiper	92-2
Removing and installing windshield wipers beginning with 1983 models	92-6

Exterior Lights, Lamps, Switches

Bulb chart	94-1
Aiming headlights	94-2
Adjusting fog lights	94-3
Removing and installing steering column switch	94-4
Removing and installing switch in instrument cover	94-5
Removing and installing sealed beam headlight units	94-6
Headlight beam regulator, checking	94-8
Headlight beam regulator, removing and installing	94-10
Main headlamps, adjusting - USA ('80 models onward)	94-14

Interior Lights

Bulb chart	96-1
Control panel variations R.o.W.	96-2
Control panel variations USA	96-3

	Page
Instruments, Fuel Gauge, Alarm System	
Diagnosis / troubleshooting alarm system	D90 - 1
Wiring	
Ground points on car	97-01
Correction to circuit diagrams, models 87 and 88, sheet 7	97-03
Additional electrical equipment	97-05
Repairing wiring harness no. 121	97-07
Current flow diagram, symbols and explanation	97-1
Current flow diagram	97-3